

Universität Freiburg – Institut für Physik

Applied Theoretical Physics

Prof. Dr. Michael Moseler

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Computational Physics: Materials Science - Exercise 2, SS 2026

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Tutorials: Mr. Andreas Doell

Problem 3. Wave packet dynamics in a double-well potential

In this exercise, we simulate the time evolution of a quantum mechanical wave packet in a symmetric double-well potential. We compare the dynamics to the trajectory of a classical particle and study quantum tunneling between the two wells.

Implementation

- Choose a smooth, symmetric one-dimensional double-well potential with two minima and a central barrier. One possible choice is a quartic potential with two minima at $\pm d$ and barrier height V_b :

$$V(x) = V_b (x^2 - d^2)^2 / d^4,$$

- Use a Gaussian function without plane-wave contribution as a wavefunction at $t = 0$:

$$\psi(x, 0) = \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^{1/2} \exp\left(-\frac{(x - x_m)^2}{4\sigma^2} \right)$$

- Initially place the wave packet into one of the wells.

Analysis

- Simulate the dynamics using the split-operator method, assuming the mass of a hydrogen atom. Generate an *interactive* plot to study the time evolution of the wavefunction. Also simulate a classical particle in the same double-well potential. Compare the quantum mechanical expectation value $\langle x \rangle(t)$ to the classical position $x(t)$. Choose parameters such that you observe tunnelling for a quantum particle but not for a classical one. Make sure your simulations are long enough to observe tunnelling.
- Plot the populations P_L, P_R of the two wells as a function of time.
- Plot the expectation values of the kinetic, potential, and total energy as a function of time. Repeat with a larger time step and compare the results.
- Run simulations with $\langle E \rangle > V_b$ and with $\langle E \rangle < V_b$ and compare the dynamics.
- Compare the dynamics obtained with the hydrogen mass to the results obtained with the mass of a carbon atom.
- Play around with the parameters. Are they physically reasonable? Feel free to explore whatever you find interesting. For example, you could modify the potential. How do the dynamics change? Graphically present your most interesting results!

What to submit

Hand in a zip file on ILIAS containing the following files:

- your Python source code
- a short report in PDF format with your main results

A good submission contains:

- A short description of the code and the goal of the exercise
- A brief description how you managed to accomplish the task. This includes the choice of parameters and everything necessary to reproduce your exact plots
- A *selection* of plots which illustrate your observations